Mathematical Foundation

For AI-related Students

Ping Li
Hangzhou Dianzi University

Linear Algebra

Scalars, Vectors, Matrices and Tensors

$s \in \mathbb{R}$

$n \in \mathbb{N}$

$\boldsymbol{x}$ $\boldsymbol{A}$ $\mathsf{\mathbf{A}}$

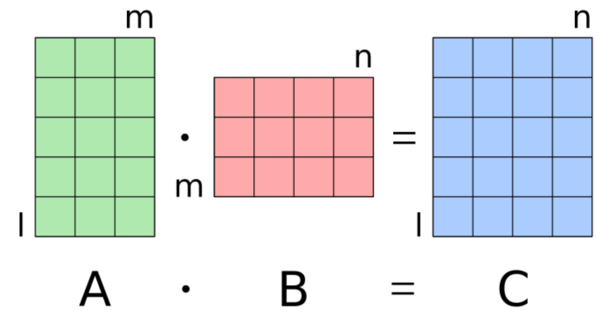
Scalar	Vector	Matrix	Tensor
1	$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	$\begin{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix} & \begin{bmatrix} 3 & 2 \end{bmatrix} \\ \begin{bmatrix} 1 & 7 \end{bmatrix} & \begin{bmatrix} 5 & 4 \end{bmatrix} \end{bmatrix}$

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \boldsymbol{x}_{-1}$$

$$\boldsymbol{A} = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{bmatrix} \Rightarrow \boldsymbol{A}^\top = \begin{bmatrix} A_{1,1} & A_{2,1} & A_{3,1} \\ A_{1,2} & A_{2,2} & A_{3,2} \end{bmatrix}$$

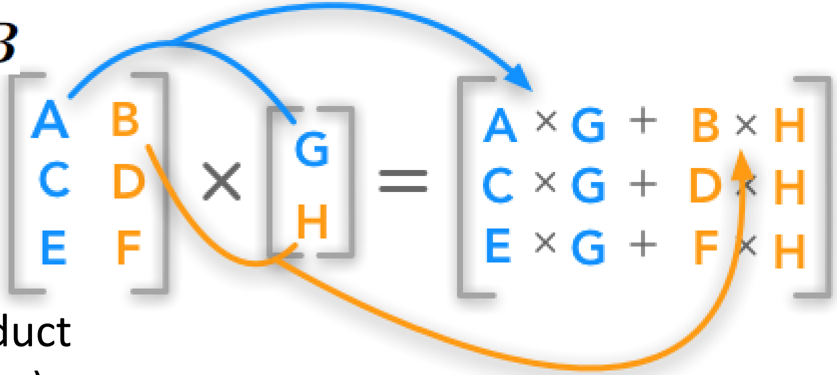
$\boldsymbol{A}_{i,:}$

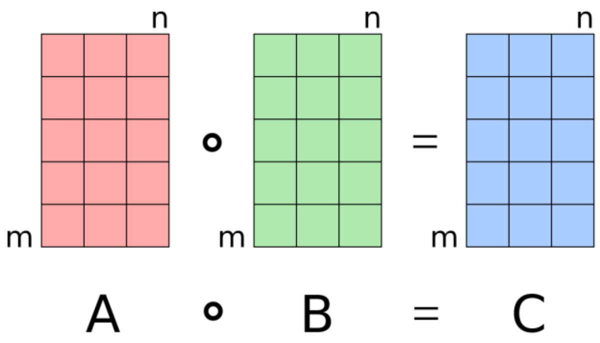
Multiplying Matrices and Vectors



$$C_{i,j} = \sum_k A_{i,k} B_{k,j}$$

$$C = AB$$





Hadmard product
(element-wise)

$$A \odot B$$

Multiplying Matrices and Vectors

$$\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \bullet \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = (u_1 * v_1) + (u_1 * v_2) + (u_3 * v_3)$$

$$\boldsymbol{x}^\top \boldsymbol{y} = \boldsymbol{y}^\top \boldsymbol{x}$$

$$\boxed{\begin{array}{ll} \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} & \boldsymbol{A} \in \mathbb{R}^{m \times n} \\ & \boldsymbol{b} \in \mathbb{R}^{m} \\ & \boldsymbol{x} \in \mathbb{R}^{n} \end{array}}$$

$$\boxed{\boldsymbol{A}_{m,1} x_1 + \boldsymbol{A}_{m,2} x_2 + \cdots + \boldsymbol{A}_{m,n} x_n = b_m}$$

Identity and Inverse Matrices

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\forall \boldsymbol{x} \in \mathbb{R}^n, \boldsymbol{I}_n \boldsymbol{x} = \boldsymbol{x}.$$

$$\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}_n$$

$$\boxed{\begin{aligned} \boldsymbol{A}\boldsymbol{x} &= \boldsymbol{b} \\ \boldsymbol{A}^{-1}\boldsymbol{A}\boldsymbol{x} &= \boldsymbol{A}^{-1}\boldsymbol{b} \\ \boldsymbol{I}_n\boldsymbol{x} &= \boldsymbol{A}^{-1}\boldsymbol{b} \end{aligned}}$$

$$\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}$$

Inverse condition

det(A) != 0, full rank square matrix, non-singular

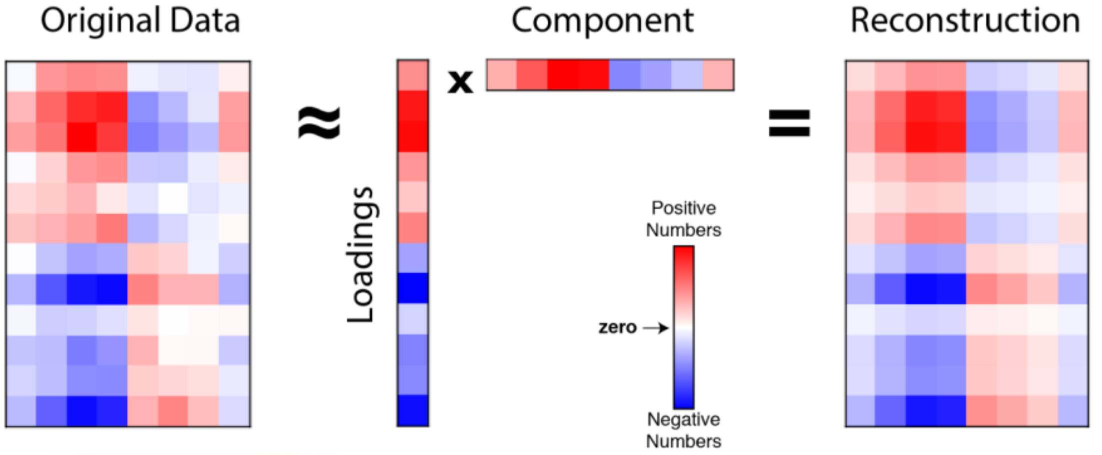
$$\begin{pmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 9 & 10 & 11 & 12 \end{pmatrix} \dashrightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

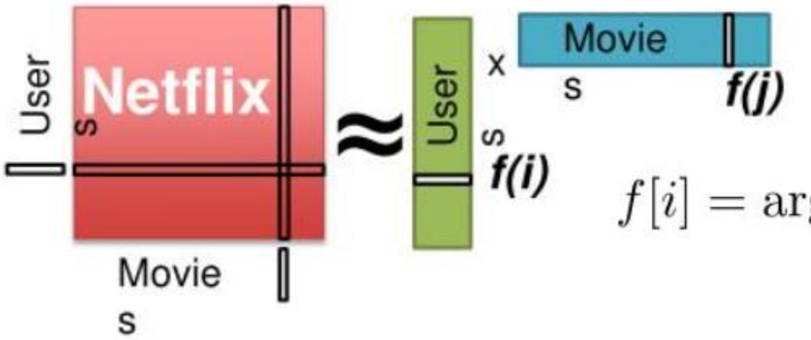
$$\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}^{-1} = \frac{1}{4x6-7x2}\begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \frac{1}{10}\begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$$

Low-rank Approximation





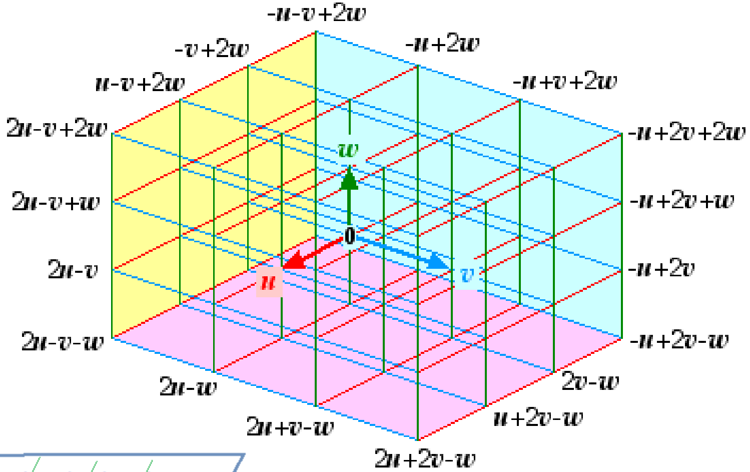
$$f[i] = \arg\min_{w \in \mathbb{R}^d} \sum_{j \in \mathrm{Nbrs}(i)} \left(r_{ij} - w^T f[j]\right)^2 + \lambda ||w||_2^2$$

Linear Dependence and Span

$$z = \alpha x + (1 - \alpha) y$$

$$Ax = \sum_i x_i A_{:,i}$$ (Linear combination)

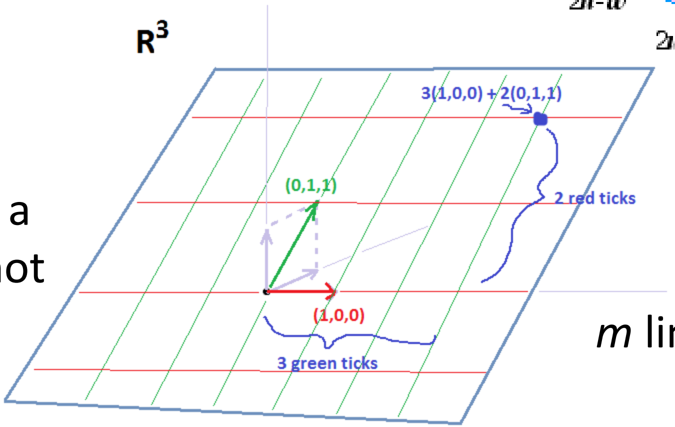
$$\sum_i c_i v^{(i)}$$ (formal)



$$b \in \mathbb{R}^m \quad n \geq m$$

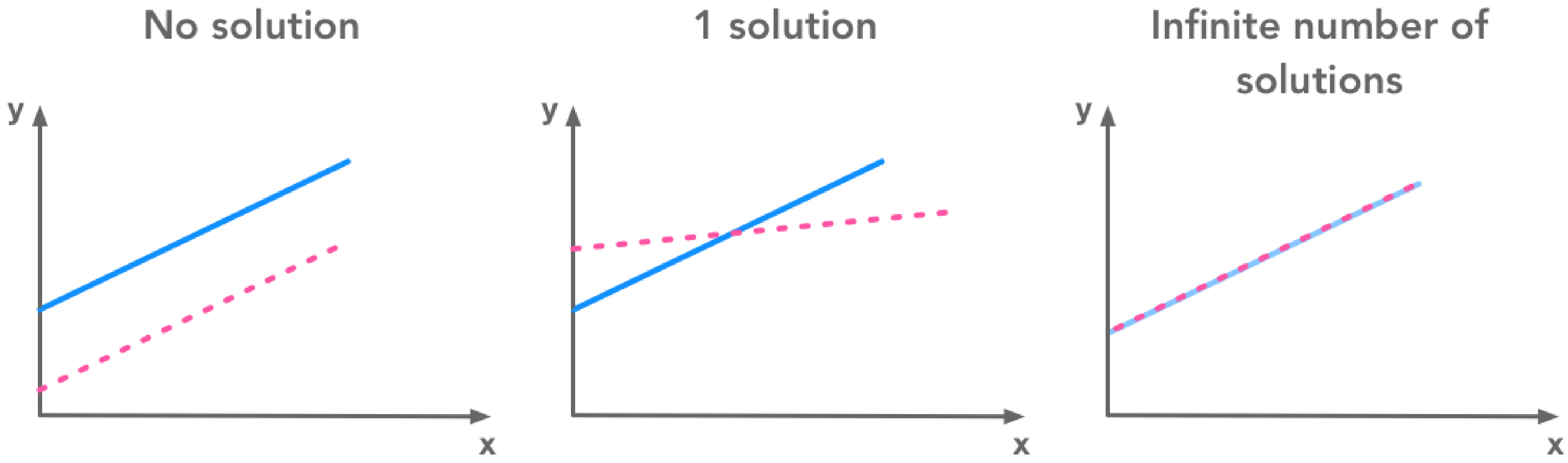
(necessary condition to have a solution at every point, but not sufficient as it may exist redundant columns)

R^3



m linearly independent columns

Linear Dependence and Span



A system of equations has no solution, 1 solution or an infinite number of solutions

Span: the set of all points obtainable by linear combination of basis vectors.

Norms

$$\|\boldsymbol{X}\|_2 := \sqrt{x_1^2 + \cdots + x_n^2}.$$

$$\|\mathbf{x}\|_\infty := \max_i |x_i|$$

$$\|\boldsymbol{x}\|_1 := \sum_{i=1}^{n} |x_i| .$$

$$\|\mathbf{x}\|_p := \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p} \quad p \in \mathbb{R}, p \geq 1$$

$$||A||_F = \sqrt{\sum_{i,j} A_{i,j}^2}$$

$$\boldsymbol{x}^\top \boldsymbol{y} = ||\boldsymbol{x}||_2 ||\boldsymbol{y}||_2 \cos\theta$$

Special Matrices and Vectors

- Diagonal matrix

$$\mathrm{diag}(\boldsymbol{v})^{-1} = \mathrm{diag}([1/v_1, \dots, 1/v_n]^\top)$$

- Symmetric matrix

$$\boldsymbol{A} = \boldsymbol{A}^\top$$

- Unit vector

$$||\boldsymbol{x}||_2 = 1$$

- Orthogonal matrix

Diagonal matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

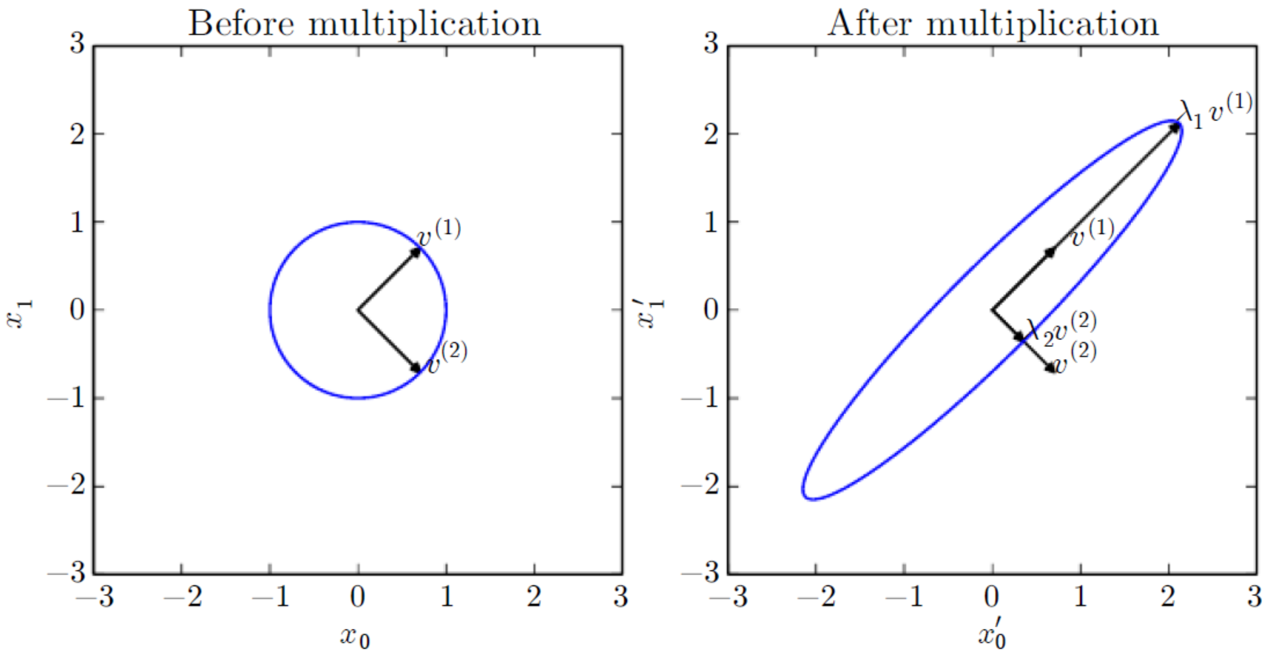
Symmetric matrix

$$\begin{bmatrix} 1 & 2 & 6 \\ 2 & 8 & 4 \\ 6 & 4 & 5 \end{bmatrix}$$

$$\boldsymbol{A}^\top \boldsymbol{A} = \boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{I} \qquad \boldsymbol{A}^{-1} = \boldsymbol{A}^\top$$

Eigen-decomposition

$$Av = \lambda v. \qquad \{\lambda_1, \ldots, \lambda_n\} \qquad A = V\,\mathrm{diag}(\boldsymbol{\lambda})V^{-1}$$



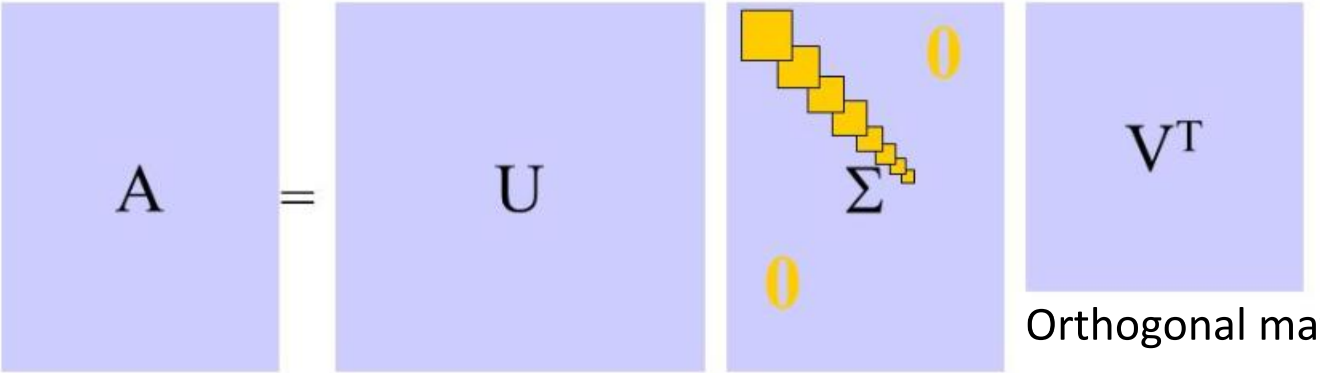
Eigen-decomposition

- Not every matrix can be decomposed into eigenvalues and eigenvectors
- It may involve complex rather than real numbers

$$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^\top$$

- Positive Semi-Definite (PSD): $\forall \boldsymbol{x},\ \boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \geq 0$

Singular Value Decomposition



Orthogonal matrix Diagonal matrix Orthogonal matrix

m x n m x m m x n n x n

$$A = V \operatorname{diag}(\boldsymbol{\lambda}) V^{-1}$$

$$A = UDV^{\top}$$

The Moore-Penrose Pseudoinverse

$$\boldsymbol{A}^{+} = \lim_{\alpha \searrow 0} (\boldsymbol{A}^{\top}\boldsymbol{A} + \alpha \boldsymbol{I})^{-1}\boldsymbol{A}^{\top}$$

$$\boldsymbol{A}^{+} = \boldsymbol{V}\boldsymbol{D}^{+}\boldsymbol{U}^{\top}$$

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{y} \Rightarrow \boldsymbol{x} = \boldsymbol{A}^{+}\boldsymbol{y}$$

The pseudo-inverse of D is obtained by taking the reciprocal of its non-zero elements then taking the transpose of the resulting matrix

The Trace Operator

$$\mathrm{Tr}(\boldsymbol{A}) = \sum_i \boldsymbol{A}_{i,i} \qquad \mathrm{Tr}(\boldsymbol{A}) = \mathrm{Tr}(\boldsymbol{A}^\top)$$

$$||A||_F = \sqrt{\mathrm{Tr}(\boldsymbol{A}\boldsymbol{A}^\top)}$$

$$\mathrm{Tr}\left(\prod_{i=1}^{n} \boldsymbol{F}^{(i)}\right) = \mathrm{Tr}\left(\boldsymbol{F}^{(n)} \prod_{i=1}^{n-1} \boldsymbol{F}^{(i)}\right)$$

$$\mathrm{Tr}(\boldsymbol{A}\boldsymbol{B}) = \mathrm{Tr}(\boldsymbol{B}\boldsymbol{A})$$

$$\boldsymbol{A} \in \mathbb{R}^{m\times n} \quad \boldsymbol{B} \in \mathbb{R}^{n \times m}$$

$$\begin{bmatrix} 0 & 2 & 3 \\ 1 & 6 & 4 \\ 2 & 6 & 1 \end{bmatrix}$$

Trace

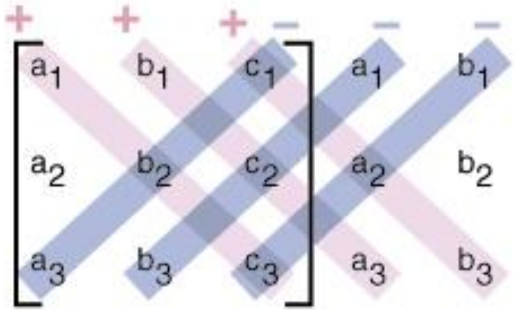
0+6+1=7

The Determinant

- $\det(\boldsymbol{A})$ function mapping matrices to real scalars
- The determinant is equal to the product of all the eigenvalues of the matrix

$$Tr(\mathbf{A}) = \sum_{j=1}^{n} \lambda_j$$

$$|\mathbf{A}| = \prod_{j=1}^{n} \lambda_j$$



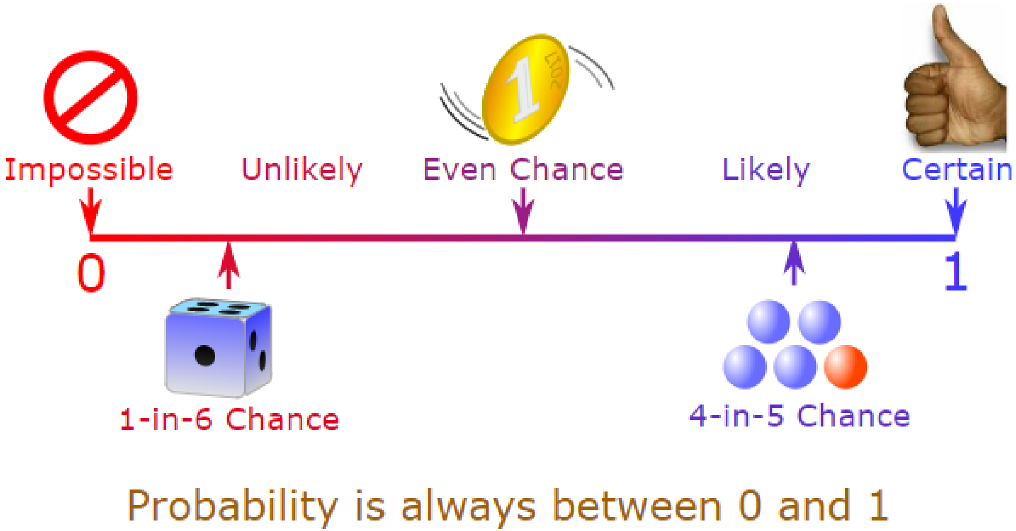
$$\det A = (a_1 b_2 c_3 + b_1 c_2 a_3 + c_1 a_2 b_3) - (a_3 b_2 c_1 + b_3 c_2 a_1 + c_3 a_2 b_1)$$

$$|A| = \frac{1}{|A^{-1}|}$$

$$|A| = |A^{-1}|$$

Probability

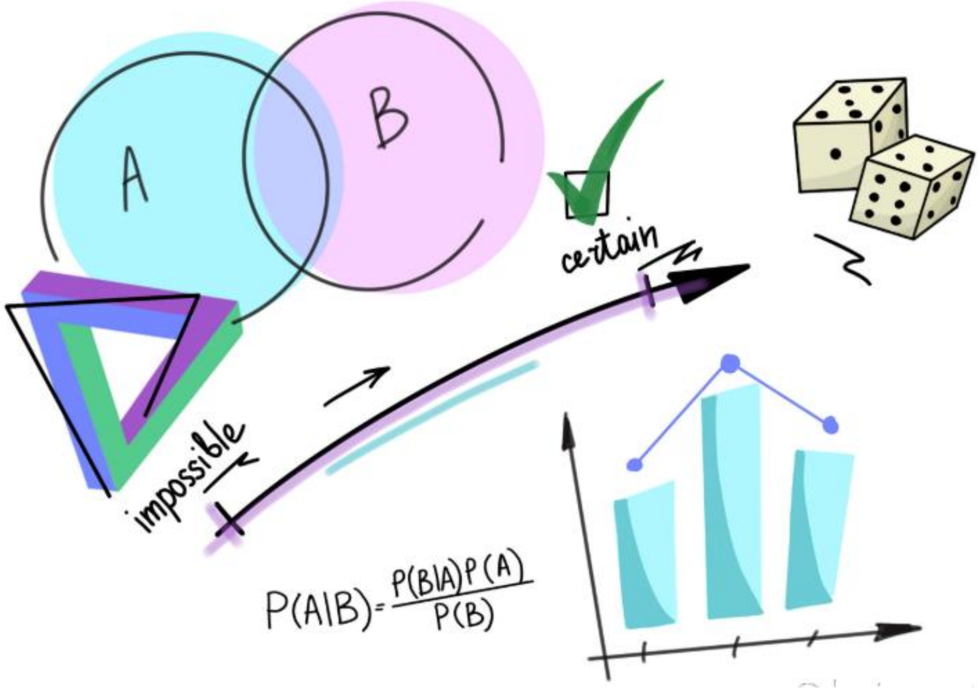
Probability Theory



- Represent uncertain statements and quantify uncertainty
- How AI systems should reason
- Analyze the behavior of AI systems

Why Probability?

Machine learning always deals with uncertain quantities and sometimes stochastic/non-deterministic quantities.



Possible Sources of Probability

- Inherent stochasticity in the system
- Incomplete observability
- Incomplete modeling

Random Variables

- Take on different values randomly
- Comply with a probability distribution
- Discrete ({0,1}) or continuous ([0,1])

Probability Distributions

- Role: how likely a random variable is to take on each of its possible states.
- Discrete variables + probability mass function(PMF)

$$P(\mathrm{x}=x),\ \ \mathrm{x}\sim P(x),\ \ P(\mathrm{x}=x,\mathrm{y}=y)$$

  - Uniform distribution: $P(\mathrm{x}=x_i)=\frac{1}{k}$
- Continuous variables + probability density function(PDF)

$$f(x)=\begin{cases}\frac{1}{b-a} & \text{for } a\le x\le b,\\ 0 & \text{for } x<a \text{ or } x>b\end{cases}$$

	Discrete		Continuous
Probability Distribution	x_1 x_2 $\cdots$ x_n	p_1 p_2 $\cdots$ p_n	pdf: f(x)
F(x)	$\sum_{i=1}^{n} p_i = 1$		$\int_{-\infty}^{\infty} f(x)dx = 1$
Mean μ	$\sum_{i=1}^{n} x_i p_i$		$\int_{-\infty}^{\infty} xf(x)dx$
Variance σ^2	$\sum_{i=1}^{n} (x_i - \mu)^2 p_i$		$\int_{-\infty}^{\infty} (x-\mu)^2 f(x)dx$

Marginal Probability

- Definition: The probability distribution over a subset of variables set (sum across row or column given P(x,y) grids)

Marginal probabilities

	Pass	Fail	Total	
Males	46	56	102	P(male) = 0.51
Females	68	30	98	P(female) = 0.49
Total	114	86	200	
	P(passed) = 0.57	P(failed = 0.43)		

$$P(\mathrm{x} = x) = \sum_y P(\mathrm{x} = x, \mathrm{y} = y)$$

$$p(x) = \int p(x, y)dy$$

Conditional Probability

- The probability of some event when other event has happened.

Probability of event A occured and event B occured

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

Probability of event A given B has occured

Probability of event B

$$P(\mathrm{y} = y \mid \mathrm{x} = x) = \frac{P(\mathrm{y} = y, \mathrm{x} = x)}{P(\mathrm{x} = x)}$$

Chain Rule

- Product rule of probability

$$P(\mathrm{x}^{(1)}, \ldots, \mathrm{x}^{(n)}) = P(\mathrm{x}^{(1)}) \Pi_{i=2}^{n} P(\mathrm{x}^{(i)} \mid \mathrm{x}^{(1)}, \ldots, \mathrm{x}^{(i-1)})$$

$$\begin{aligned}
P(\mathrm{a}, \mathrm{b}, \mathrm{c}) &= P(\mathrm{a} \mid \mathrm{b}, \mathrm{c}) P(\mathrm{b}, \mathrm{c}) \\
P(\mathrm{b}, \mathrm{c}) &= P(\mathrm{b} \mid \mathrm{c}) P(\mathrm{c}) \\
P(\mathrm{a}, \mathrm{b}, \mathrm{c}) &= P(\mathrm{a} \mid \mathrm{b}, \mathrm{c}) P(\mathrm{b} \mid \mathrm{c}) P(\mathrm{c})
\end{aligned}$$

Independence

$$\forall x \in \mathrm{x}, y \in \mathrm{y},\ p(\mathrm{x} = x, \mathrm{y} = y) = p(\mathrm{x} = x)p(\mathrm{y} = y)$$

Conditional Independence

$$\forall x \in \mathrm{x}, y \in \mathrm{y}, z \in \mathrm{z},\ p(\mathrm{x} = x, \mathrm{y} = y \mid \mathrm{z} = z) = p(\mathrm{x} = x \mid \mathrm{z} = z)p(\mathrm{y} = y \mid \mathrm{z} = z)$$

Expectation and Variance

Expectation is the average or mean value that some function f(x) takes on when *x* is drawn from its probability distribution P(x).

$$\mathbb{E}_{\mathrm{x}\sim P}[f(x)] = \sum_x P(x)f(x) \qquad \mathbb{E}_{\mathrm{x}\sim p}[f(x)] = \int p(x)f(x)dx$$

Variance gives a measure of how much the values of a function of a random variable vary as we sample different values from its probability distribution.

$$\mathrm{Var}(f(x)) = \mathbb{E}\left[(f(x) - \mathbb{E}[f(x)])^2\right]$$

Covariance

Covariance gives some sense of how much two values are linearly related to each other as well as the scale of these variables.

$$\mathrm{Cov}(f(x), g(y)) = \mathbb{E}\left[(f(x) - \mathbb{E}\left[f(x)\right])\left(g(y) - \mathbb{E}\left[g(y)\right]\right)\right]$$

Covariance Matrix

$$\boldsymbol{x} \in \mathbb{R}^n \qquad \mathrm{Cov}(\mathbf{x})_{i,j} = \mathrm{Cov}(\mathrm{x}_i, \mathrm{x}_j)$$

$$\mathrm{Cov}(\mathrm{x}_i, \mathrm{x}_i) = \mathrm{Var}(\mathrm{x}_i)$$

Common Probability Distributions

- Bernoulli distribution

$$P(\mathrm{x} = 1) = \phi$$

$$P(\mathrm{x} = 0) = 1 - \phi$$

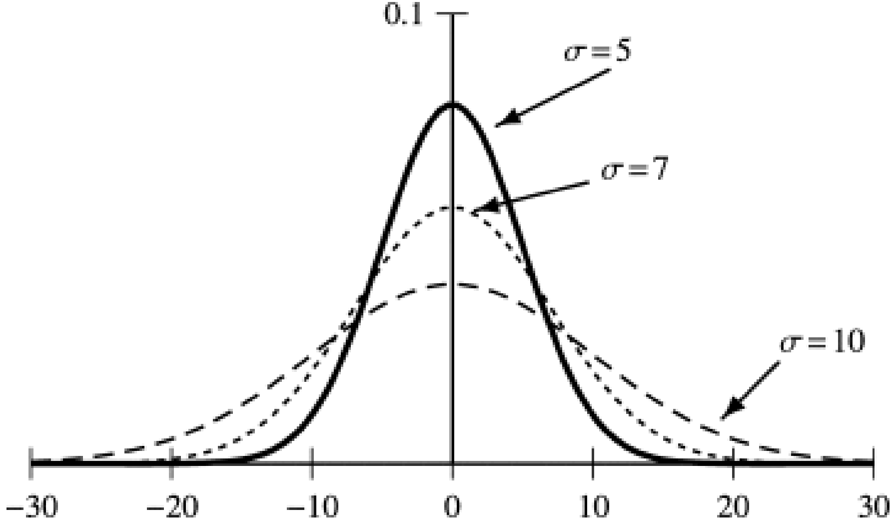
$$P(\mathrm{x} = x) = \phi^x (1 - \phi)^{1-x}$$

- Multinoulli / Categorical distribution
  - It is a distribution over a single discrete variable with *k* different states.

$$\boldsymbol{p} \in [0, 1]^{k-1}$$

- Gaussian distribution

$$\mathcal{N}(x;\mu,\sigma^2)=\sqrt{\frac{1}{2\pi\sigma^2}}\exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right)$$



- Multivariate normal distribution $\mathbb{R}^n$

$$\mathcal{N}(\boldsymbol{x};\boldsymbol{\mu},\boldsymbol{\Sigma}) = \sqrt{\frac{1}{(2\pi)^n \det(\boldsymbol{\Sigma})}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right)$$

Precise Matrix to evaluate PDF

$$\mathcal{N}(\boldsymbol{x};\boldsymbol{\mu},\boldsymbol{\beta}^{-1}) = \sqrt{\frac{\det(\boldsymbol{\beta})}{(2\pi)^n}} \exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\beta}(\boldsymbol{x}-\boldsymbol{\mu})\right)$$

Exponential and Laplace Distributions

$$p(x;\lambda) = \lambda \boxed{\mathbf{1}_{x\geq 0}} \exp\left(-\lambda x\right)$$

$$\mathrm{Laplace}(x;\mu,\gamma) = \frac{1}{2\gamma}\exp\left(-\frac{\boxed{|x-\mu|}}{\gamma}\right)$$

Dirac / Empirical Distributions

$$p(x) = \delta(x - \mu)$$

$$\hat{p}(\boldsymbol{x}) = \frac{1}{m} \sum_{i=1}^{m} \delta(\boldsymbol{x} - \boldsymbol{x}^{(i)})$$

It is the probability density that maximizes the likelihood of the training data.

Mixtures of Distributions

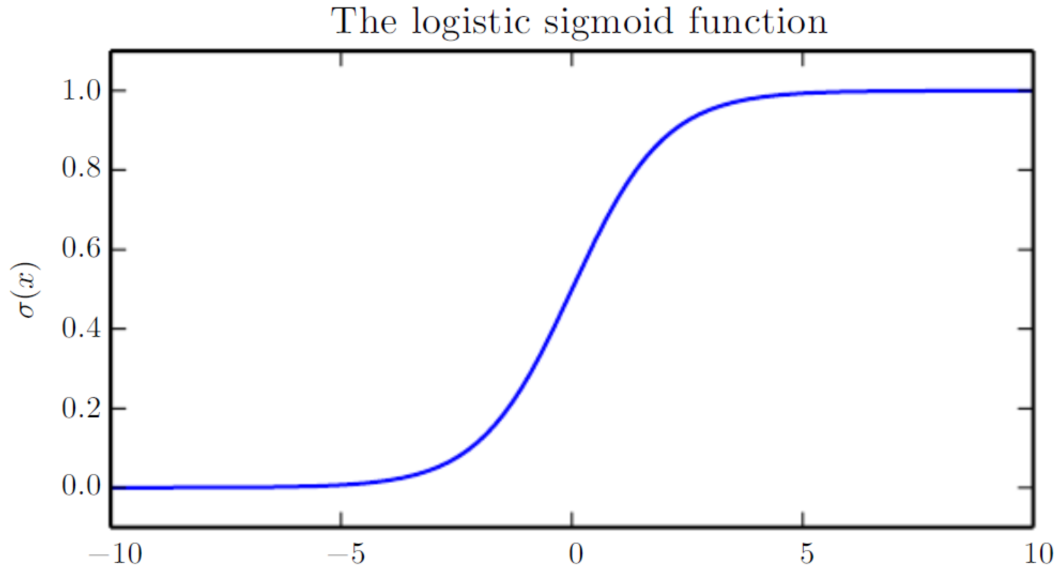
- Sampling a component identity from a multinoulli distribution.

$$P(\mathrm{x}) = \sum_i P(\mathrm{c} = i) P(\mathrm{x} \mid \mathrm{c} = i)$$

- Latent variable: a random variable cannot be observed directly
- Gaussian mixture model: $p(\mathbf{x} \mid \mathrm{c} = i)$ are Gaussians
- Prior probability: $\alpha_i = P(\mathrm{c} = i)$ for each component *I*
- Posterior probability: $P(\mathrm{c} \mid \boldsymbol{x})$

Useful Properties of Common Functions

- Logistic sigmoid:



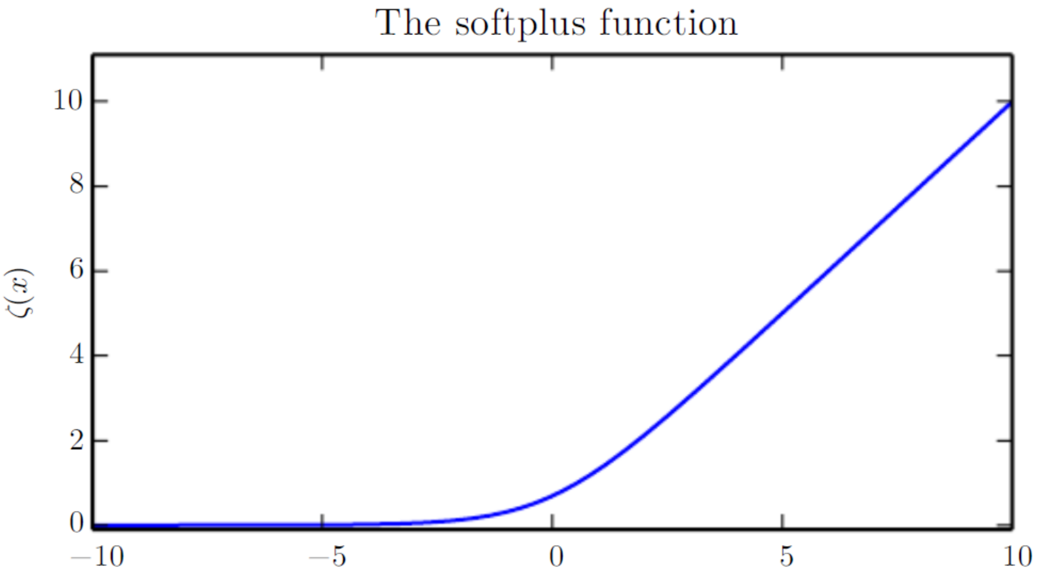
$$\sigma(x) = \frac{1}{1 + \exp(-x)}$$

$$\sigma(x) = \frac{\exp(x)}{\exp(x) + \exp(0)}$$

$$\frac{d}{dx}\sigma(x) = \sigma(x)(1 - \sigma(x))$$

Useful Properties of Common Functions

- Softplus function:



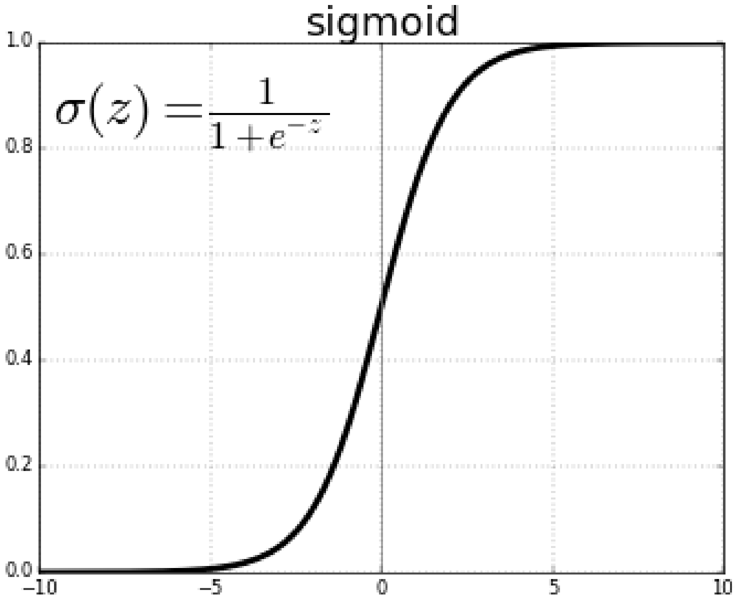
$$\zeta(x) = \log(1 + \exp(x))$$

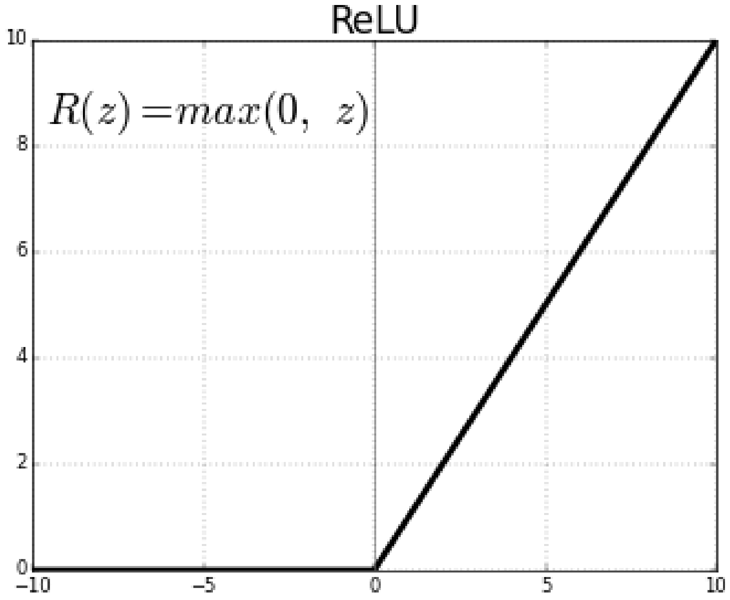
$$x^+ = \max(0, x)$$

$$\frac{d}{dx}\zeta(x) = \sigma(x)$$

$$\zeta(x) - \zeta(-x) = x$$

Activation Functions





Bayes' Rule

$$P(\mathrm{x} \mid \mathrm{y}) = \frac{P(\mathrm{x})P(\mathrm{y} \mid \mathrm{x})}{P(\mathrm{y})}$$

$$P(\mathrm{y}) = \sum_x P(\mathrm{y} \mid x)P(x)$$

Optimization

Optimization Problem

- Minimization problem: $\min_{x \in D} f(x)$
- Minimizer of f(x): $x^* = \arg\min_{x \in D} f(x)$

$$\min_{x \in \mathbb{R}} \sin(x) \Rightarrow -\frac{\pi}{2} + 2n\pi$$

- Maximization problem: $\min_{x \in D} -f(x) \;\Big|\; \max_{x \in D} f(x)$

Local minimum

- Some x that leads to the smallest objective value in its local neighborhood

$$\|\boldsymbol{x} - \boldsymbol{x}^\star\| \leq r \qquad f(\boldsymbol{x}^\star) \leq f(\boldsymbol{x}) \qquad \text{radius } r > 0$$

- Points with an all-zero gradient are also called stationary points or critical points. (e.g., maximizer, minimizer, saddle point)

Gradient

$$\left[\frac{\partial \boldsymbol{x}}{\partial y}\right]_i = \frac{\partial x_i}{\partial y}$$

$$\left[\frac{\partial x}{\partial \boldsymbol{y}}\right]_i = \frac{\partial x}{\partial y_i}$$

$$\left[\frac{\partial \boldsymbol{x}}{\partial \boldsymbol{y}}\right]_{ij} = \frac{\partial x_i}{\partial y_j}$$

$$\left[\frac{\partial y}{\partial X}\right]_{ij} = \frac{\partial y}{\partial x_{ij}}$$

$$\frac{\partial \boldsymbol{x}^T \boldsymbol{y}}{\partial \boldsymbol{x}} = \boldsymbol{y}$$

$$\frac{\partial \boldsymbol{a}^T X \boldsymbol{b}}{\partial X} = \boldsymbol{a}\boldsymbol{b}^T$$

The Matrix Cookbook

https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf

Convex and concave optimization

- In convex minimization, any local minimum must be a global minimum.

for any $\boldsymbol{x} \in S$, $\boldsymbol{y} \in S$ and $0 \leq \lambda \leq 1$

$$\lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y} \in S$$

$S \subseteq \mathbb{R}^d$; ← Convex set

Jensen's inequality

$$f\left(\sum_{i=1}^{n} w_i \boldsymbol{x}_i\right) \leq \sum_{i=1}^{n} w_i f(\boldsymbol{x}_i)$$

$$f(\lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y}) \leq \lambda f(\boldsymbol{x}) + (1 - \lambda) f(\boldsymbol{y})$$ ← Convex function $f(x) = x^2$

* A twice differentiable function is convex (concave) if its second derivative is non-negative (non-positive). e.g. **ln(x)**, **x^4**

Constrained Optimization

$$\begin{aligned} \min \quad & f(\boldsymbol{x}) \\ \text{s.t.} \quad & g_1(\boldsymbol{x}) = 0, \\ & \dots \\ & g_m(\boldsymbol{x}) = 0\,. \end{aligned}$$

$$L(\boldsymbol{x}, \boldsymbol{\lambda}) = f(\boldsymbol{x}) - \boldsymbol{\lambda}^T \boldsymbol{g}(\boldsymbol{x})$$

Lagrange function

$$\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \dots, \lambda_m)^T$$

Unconstrained optimization objective

$$\begin{aligned} \min \quad & f(\boldsymbol{x}) = \boldsymbol{v}^T \boldsymbol{x} \\ \text{s.t.} \quad & \boldsymbol{x}^T \boldsymbol{x} = 1 \end{aligned}$$

$$L(\boldsymbol{x}, \lambda) = \boldsymbol{v}^T \boldsymbol{x} - \lambda(\boldsymbol{x}^T \boldsymbol{x} - 1)$$

$$\boldsymbol{v} = 2\lambda \boldsymbol{x}$$

$$\|\boldsymbol{v}\|^2 = \boldsymbol{v}^T \boldsymbol{v} = 4\lambda^2 \boldsymbol{x}^T \boldsymbol{x} = 4\lambda^2$$